Integrals — Revision Summary

Indefinite, definite, techniques, inequalities, and limit interchange

1 Indefinite integrals — the antiderivative

Theorem (Definition). *F is an* antiderivative *of f on an interval I if $F'(x) = f(x)$ for all $x \in I$. We write $\int f(x)\,dx = F(x) + C$.*

Why "$+C$". If F is one antiderivative, so is $F + C$ for any constant C (since $(F + C)' = F'$). Conversely, by MVT, two antiderivatives on a connected interval differ by a constant. So the indefinite integral is a *family of curves*, all vertical translates of each other — the constant disappears the moment you take a derivative.

Caveat on disconnected domains. On a disjoint union of intervals, the constants on each piece can be different. For example, $\int dx/x = \ln|x| + C$, but the C on $x > 0$ and on $x < 0$ are independent.

1.1 Standard antiderivatives

(with $+C$ everywhere)

$\int x^n\,dx = \dfrac{x^{n+1}}{n+1}$ $(n \neq -1)$	$\int \dfrac{dx}{x} = \ln|x|$	$\int e^x\,dx = e^x$
$\int \sin x\,dx = -\cos x$	$\int \cos x\,dx = \sin x$	$\int \sec^2 x\,dx = \tan x$
$\int \tan x\,dx = -\ln|\cos x|$	$\int \sec x\,dx = \ln|\sec x + \tan x|$	$\int \dfrac{dx}{1+x^2} = \arctan x$
$\int \dfrac{dx}{\sqrt{1-x^2}} = \arcsin x$	$\int \dfrac{dx}{\sqrt{x^2 \pm a^2}} = \ln|x + \sqrt{x^2 \pm a^2}|$	$\int a^x\,dx = \dfrac{a^x}{\ln a}$

1.2 Core techniques

Substitution (u-sub). If $u = g(x)$, then $\int f(g(x))g'(x)\,dx = \int f(u)\,du$. The chain rule run backwards. Trigger: spot a function and its derivative side-by-side.

Integration by parts. $\int u\,dv = uv - \int v\,du$. Choose u to simplify on differentiation, dv to be readily integrable. The mnemonic "LIATE" (Logarithmic, Inverse trig, Algebraic, Trig, Exponential) gives a usable priority order for picking u.

Partial fractions. For rational integrands $P(x)/Q(x)$ with $\deg P < \deg Q$: factor Q into linear and irreducible quadratic factors, decompose, and integrate term by term. For repeated factors $(x-a)^k$: include terms $A_1/(x-a) + A_2/(x-a)^2 + \cdots + A_k/(x-a)^k$.

Trigonometric substitutions.

- $\sqrt{a^2 - x^2}$: $x = a\sin\theta$.
- $\sqrt{a^2 + x^2}$: $x = a\tan\theta$.
- $\sqrt{x^2 - a^2}$: $x = a\sec\theta$.

Weierstrass substitution. For rational functions of sin and cos: $t = \tan(x/2)$, giving $\sin x = \dfrac{2t}{1+t^2}$, $\cos x = \dfrac{1-t^2}{1+t^2}$, $dx = \dfrac{2\,dt}{1+t^2}$.

2 Complex-method tricks (Euler's formula)

When integrands involve $e^{ax}\cos bx$, $e^{ax}\sin bx$, $e^{\cos x}\cos(\sin x)$, or other "mixed" expressions, replace trig functions by their complex-exponential form $e^{ix} = \cos x + i\sin x$ and take real or imaginary parts at the end.

Standard application: $\int e^{ax}\cos(bx)\,dx$. Write $\cos bx = \operatorname{Re} e^{ibx}$:

$$\int e^{ax}\cos(bx)\,dx = \operatorname{Re}\int e^{(a+ib)x}\,dx = \operatorname{Re}\frac{e^{(a+ib)x}}{a+ib} = \frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2} + C.$$

No integration by parts, no recursive trick — one line.

Showcase: $e^{\cos x}\cos(\sin x)$. Notice that

$$e^{\cos x}\cos(\sin x) = \operatorname{Re} e^{\cos x + i\sin x} = \operatorname{Re} e^{e^{ix}}.$$

So

$$\int_0^{2\pi} e^{\cos x}\cos(\sin x)\,dx = \operatorname{Re}\int_0^{2\pi} e^{e^{ix}}\,dx.$$

Expand $e^{e^{ix}} = \sum_{n=0}^{\infty} \frac{e^{inx}}{n!}$ and integrate term by term. For $n \geq 1$, $\int_0^{2\pi} e^{inx}\,dx = 0$. For $n = 0$, $\int_0^{2\pi} 1\,dx = 2\pi$. Hence the integral is 2π.

Pattern. For any integral of the form $\int g(e^{ix}) \cdot [\text{trig}]\,dx$ over a full period, expand g in Taylor series and use the orthogonality $\int_0^{2\pi} e^{inx}\,dx = 2\pi\delta_{n,0}$.

3 Definite integrals — Riemann sums

Theorem (Riemann integral)**.** *A bounded f on $[a, b]$ is* Riemann integrable *if upper and lower Riemann sums (over partitions of $[a, b]$) converge to a common limit as the mesh $\to 0$. We write $\int_a^b f(x)\,dx$ for this common value.*

Continuity is sufficient but not necessary. Every continuous function on $[a, b]$ is Riemann integrable. So is every monotone function (even with countably many jumps), and every bounded function with only countably many discontinuities.

When discontinuities are "small". A bounded f on $[a, b]$ is Riemann integrable iff its set of discontinuities is small in a precise sense (technically: of *measure zero*). Practically: countably many discontinuities are fine, jumping at every irrational point can still be fine (Thomae), but discontinuity at every point fails (Dirichlet).

3.1 Fundamental Theorem of Calculus

Theorem (FTC, Part I — evaluation)**.** *If f is continuous on $[a, b]$ and F is any antiderivative of f, then*

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Theorem (FTC, Part II — differentiation under integration)**.** *If f is continuous on $[a, b]$ and $G(x) = \int_a^x f(t)\,dt$, then G is differentiable on (a, b) and $G'(x) = f(x)$.*

4 Symmetry and reflection tricks

The $f(x) \to f(a + b - x)$ trick. For any integrable f on $[a, b]$:

$$\int_a^b f(x)\,dx = \int_a^b f(a + b - x)\,dx.$$

(Substitute $u = a + b - x$.) Adding the two forms often produces a clean answer.

Classical example. $I = \displaystyle\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x}\,dx$. Apply $x \mapsto \pi/2 - x$:

$$I = \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x}\,dx, \quad \text{so} \quad 2I = \int_0^{\pi/2} 1\,dx = \frac{\pi}{2}, \quad I = \frac{\pi}{4}.$$

Even/odd symmetry on $[-a, a]$.

$$\int_{-a}^{a} f(x)\,dx = \begin{cases} 2\int_0^a f(x)\,dx & \text{if } f \text{ is even,} \\ 0 & \text{if } f \text{ is odd.} \end{cases}$$

Periodicity. If f has period T, then $\int_a^{a+T} f = \int_0^T f$ for any a. Useful to shift the interval to where the integrand is simplest.

Subinterval reflection. On $[0, a]$: $\int_0^a f(x)\,dx = \int_0^a f(a - x)\,dx$. On $[a, b]$: $\int_a^b f(x)\,dx = \int_a^b f(a + b - x)\,dx$.

5 Riemann sums as limits of sequences

The definite integral is the limit of Riemann sums; reading this backwards converts certain sequence-limits into integrals.

Theorem (Riemann-sum limit)**.** *For continuous f on $[0,1]$:*

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n} f\left(\frac{k}{n}\right)=\int_0^1 f(x)\,dx.$$

General form on $[a,b]$.

$$\lim_{n\to\infty}\frac{b-a}{n}\sum_{k=1}^{n} f\left(a+\frac{k(b-a)}{n}\right)=\int_a^b f(x)\,dx.$$

Standard examples.

- $\displaystyle\lim_{n\to\infty}\sum_{k=1}^{n}\frac{1}{n+k}=\lim\frac{1}{n}\sum\frac{1}{1+k/n}=\int_0^1\frac{dx}{1+x}=\ln 2.$
- $\displaystyle\lim_{n\to\infty}\frac{1}{n}\sqrt[n]{(n+1)(n+2)\cdots(2n)}=\exp\left(\int_0^1\ln(1+x)\,dx\right)=e^{2\ln 2-1}=\frac{4}{e}.$ (Take logs first.)
- $\displaystyle\lim_{n\to\infty}\sum_{k=1}^{n}\frac{n}{n^2+k^2}=\int_0^1\frac{dx}{1+x^2}=\frac{\pi}{4}.$

Spotting the pattern. Look for $\frac{1}{n}\sum f(k/n)$ structure. Constants like $1/n$ outside, k/n inside the function.

6 Differentiation tricks for integrals

A family of closely-related techniques: differentiate an integral with respect to its limits, with respect to a parameter inside the integrand, or apply L'Hôpital using FTC. All share the idea *differentiating an integral simplifies it.*

6.1 Leibniz integral rule (moving limits + parameter)

Theorem (Leibniz integral rule, general form)**.** *If $G(x)=\int_{u(x)}^{v(x)} f(t,x)\,dt$ with f and $\partial f/\partial x$ continuous and u,v differentiable, then*

$$G'(x)=f(v(x),x)\,v'(x)-f(u(x),x)\,u'(x)+\int_{u(x)}^{v(x)}\frac{\partial f}{\partial x}(t,x)\,dt.$$

Three contributions. The first term comes from the upper limit moving (creates new "area" at the right edge); the second from the lower limit moving (loses area at the left edge); the third from the integrand changing as x changes.

6.2 Special cases worth memorising

Pure FTC (constant limits, integrand depends on x only via the upper limit). If $G(x)=\int_a^x f(t)\,dt$:

$$G'(x)=f(x).$$

Moving upper limit only. If $G(x)=\int_a^{v(x)} f(t)\,dt$:

$$G'(x)=f(v(x))\,v'(x).$$

Pure parametric (constant limits, integrand depends on parameter). If $F(t)=\int_a^b f(x,t)\,dx$ with $f,\partial f/\partial t$ continuous:

$$F'(t)=\int_a^b\frac{\partial f}{\partial t}(x,t)\,dx.$$

This is what is usually meant by "differentiating under the integral sign" or "Feynman's trick."

6.3 Showcase 1: $\int_0^1\frac{x^a-1}{\ln x}\,dx=\ln(a+1)$ for $a>-1$

The integrand has a $\ln x$ in the denominator that resists every standard technique. The trick: introduce a as a parameter and use parametric Leibniz. Define

$$I(a)=\int_0^1\frac{x^a-1}{\ln x}\,dx.$$

Then $I(0) = 0$ (integrand is identically 0). Differentiate under the integral:

$$I'(a) = \int_0^1 \frac{\partial}{\partial a}\left(\frac{x^a - 1}{\ln x}\right) dx = \int_0^1 \frac{x^a \ln x}{\ln x}\, dx = \int_0^1 x^a\, dx = \frac{1}{a+1}.$$

Integrate back: $I(a) = \int_0^a \dfrac{dt}{t+1} = \ln(a+1)$.

Why differentiation tames it. Differentiating x^a with respect to a gives $x^a \ln x$ — which cancels the troublesome $\ln x$ in the denominator. The general lesson: when an integrand has an awkward factor that resists antidifferentiation, look for a parameter whose derivative will cancel it.

Other classics handled this way.

- $\int_0^\infty \dfrac{\sin x}{x}\, dx = \dfrac{\pi}{2}$ (introduce e^{-tx}, differentiate in t).
- $\int_0^\infty e^{-x^2}\, dx = \dfrac{\sqrt{\pi}}{2}$ (Gaussian integral; square it and switch to polar).
- $\int_0^1 \dfrac{\ln(1+x)}{1+x^2}\, dx = \dfrac{\pi}{8}\ln 2$ (parameter inserted in ln).

6.4 Showcase 2: L'Hôpital + FTC

Integrals depending on a moving limit often produce indeterminate forms when the limit tends to a boundary. L'Hôpital + FTC is the standard tool, exploiting the same "derivative of an integral is the integrand" idea.

Pattern. Limits like $\displaystyle\lim_{x\to 0} \frac{1}{x^k}\int_0^x f(t)\, dt$ become $\frac{0}{0}$. Apply L'Hôpital: the derivative of the numerator is $f(x)$ (by FTC), the derivative of the denominator is kx^{k-1}:

$$\lim_{x\to 0} \frac{1}{x^k}\int_0^x f(t)\, dt = \lim_{x\to 0} \frac{f(x)}{kx^{k-1}}.$$

Evaluate, or apply L'Hôpital again if still indeterminate.

Standard example. $\displaystyle\lim_{x\to 0} \frac{1}{x^3}\int_0^x \sin(t^2)\, dt$. By L'Hôpital: $\lim \dfrac{\sin(x^2)}{3x^2} = \dfrac{1}{3}$ (using $\sin u \sim u$).

Two applications of L'Hôpital. If after the first application the derivative-ratio is still $\frac{0}{0}$, repeat. Each FTC differentiation strips one integration:

$$\lim_{x\to 0} \frac{\int_0^x \int_0^t f(s)\, ds\, dt}{x^2} = \lim \frac{\int_0^x f(s)\, ds}{2x} = \lim \frac{f(x)}{2} = \frac{f(0)}{2}.$$

7 Area under a curve and the geometric meaning

For non-negative f, $\int_a^b f(x)\, dx$ is the area between the graph of f, the x-axis, and the vertical lines $x = a, x = b$. For sign-changing f, areas above the axis count positively, below negatively.

Total geometric area. If you want *geometric* area (everything positive), integrate $|f|$, splitting at the zeros: $\text{Area} = \int_a^b |f(x)|\, dx$.

Area between two curves. For $f \geq g$ on $[a, b]$:

$$\text{Area} = \int_a^b [f(x) - g(x)]\, dx.$$

For curves crossing, split at intersection points and take $|f - g|$.

Volume of revolution.

- **Disk/washer (around x-axis):** $V = \pi \int_a^b f(x)^2\, dx$ (or $\pi \int [f^2 - g^2]\, dx$ for washers).
- **Cylindrical shells (around y-axis):** $V = 2\pi \int_a^b x f(x)\, dx$.

Arc length. For $y = f(x)$ on $[a, b]$: $L = \int_a^b \sqrt{1 + f'(x)^2}\, dx$.

8 Mean Value Theorem for integrals

Theorem (MVT for integrals)**.** *If f is continuous on $[a, b]$, there exists $c \in [a, b]$ such that*

$$\int_a^b f(x)\, dx = f(c)(b - a).$$

Geometric meaning. The area under the graph equals the area of a rectangle with base $(b-a)$ and height $f(c)$ for some c — the function attains its *average value* somewhere on the interval.

Average value. Define $\bar{f} = \dfrac{1}{b-a}\int_a^b f(x)\,dx$. The MVT for integrals says $f(c) = \bar{f}$ for some $c \in [a,b]$.

Theorem (Generalised MVT for integrals)**.** *If f is continuous and $g \geq 0$ is integrable on $[a,b]$, there exists $c \in [a,b]$ such that*

$$\int_a^b f(x)g(x)\,dx = f(c)\int_a^b g(x)\,dx.$$

9 Improper integrals

An integral is *improper* if either the interval is unbounded or the integrand is unbounded on the interval. Define by limits:

$$\int_a^\infty f = \lim_{R\to\infty}\int_a^R f, \qquad \int_a^b f = \lim_{\varepsilon\to 0^+}\int_{a+\varepsilon}^b f \quad \text{(if } f \text{ singular at } a\text{)}.$$

Convergence tests. Mostly parallel the tests for series.

- **p-test at ∞:** $\int_1^\infty x^{-p}\,dx$ converges iff $p > 1$.
- **p-test at 0:** $\int_0^1 x^{-p}\,dx$ converges iff $p < 1$.
- **Comparison test:** If $0 \leq f \leq g$ and $\int g$ converges, so does $\int f$.
- **Limit comparison:** $\lim \dfrac{f}{g} = L \in (0,\infty)$: both converge or both diverge.
- **Absolute convergence:** $\int |f|$ converges $\Rightarrow \int f$ converges.

10 Young's inequality and inverse functions

Theorem (Young's inequality for inverse functions)**.** *Let $f : [0,\infty) \to [0,\infty)$ be continuous, strictly increasing, with $f(0) = 0$. Let $g = f^{-1}$. Then for all $a, b \geq 0$:*

$$ab \leq \int_0^a f(x)\,dx + \int_0^b g(y)\,dy.$$

Equality iff $b = f(a)$.

Geometric proof at a glance. The right-hand side is the area between the y-axis and the curve $y = f(x)$, plus the area between the x-axis and the curve. These two areas together cover the rectangle $[0,a] \times [0,b]$ (and possibly more), so the rectangle's area ab is bounded above.

Famous specialisation: Young's inequality on exponents. Take $f(x) = x^{p-1}$ for $p > 1$, so $g(y) = y^{1/(p-1)}$. With $1/p + 1/q = 1$:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

This is the bedrock under Hölder's inequality and Minkowski's inequality.

Specialisation: AM–GM via $f(x) = e^x$. ... etc. (Various other inequalities follow by choosing different f.)

11 Integral inequalities

11.1 Basic area comparison

If $f \leq g$ on $[a,b]$, then $\int_a^b f \leq \int_a^b g$. Trivial but constantly useful: bounding the integrand at every point bounds the integral.

Application. For $f \in C[a,b]$: $\left|\int_a^b f\right| \leq \int_a^b |f| \leq (b-a)\max_{[a,b]}|f|$.

11.2 Cauchy–Schwarz inequality (integral form)

Theorem (Cauchy–Schwarz)**.** *For f, g integrable on $[a,b]$:*

$$\left(\int_a^b f(x)g(x)\,dx\right)^2 \leq \left(\int_a^b f(x)^2\,dx\right)\left(\int_a^b g(x)^2\,dx\right).$$

Equality iff f and g are linearly dependent.

Proof idea. Consider $\int (f+tg)^2 \geq 0$ as a quadratic in t; its discriminant is non-positive.

Applications.

- Bound $\int fg$ in terms of $\int f^2$ and $\int g^2$ when fg has no closed-form antiderivative.
- Take $g \equiv 1$: $\left(\int_a^b f\right)^2 \le (b-a)\int_a^b f^2$. So the "square of the average" is bounded by the "average of the squares."
- Bounds on integrals of products of trig functions, e.g., $\left(\int_0^\pi \sin x \cdot f(x)\,dx\right)^2 \le \frac{\pi}{2}\int_0^\pi f(x)^2\,dx$.

11.3 Chebyshev's sum inequality (integral form)

Theorem (Chebyshev's integral inequality)**.** *If f, g are both monotonically increasing (or both decreasing) on $[a,b]$:*

$$(b-a)\int_a^b f(x)g(x)\,dx \ge \left(\int_a^b f(x)\,dx\right)\left(\int_a^b g(x)\,dx\right).$$

The inequality reverses if f is increasing while g is decreasing.

Intuition. "Same-monotonicity" functions are positively correlated — their integral product exceeds the product of integrals. "Opposite-monotonicity" functions are negatively correlated.

Application. Comparing $\int fg$ to $(\int f)(\int g)/(b-a)$ is the integral analogue of comparing covariance to zero.

11.4 Hermite–Hadamard inequality

Theorem (Hermite–Hadamard)**.** *If $f : [a,b] \to \mathbb{R}$ is convex:*

$$f\left(\frac{a+b}{2}\right) \le \frac{1}{b-a}\int_a^b f(x)\,dx \le \frac{f(a)+f(b)}{2}.$$

Reverse for concave f.

Geometric meaning. The average value of a convex function is sandwiched between the value at the midpoint (tangent below) and the average of the endpoint values (chord above). The midpoint value is a *lower* estimate of the integral, the chord average an *upper* one — this is also why the trapezoidal rule overestimates and the midpoint rule underestimates a convex function.

Application. Quick numerical bounds for integrals of convex/concave functions without computing the integral. Example: $\int_0^1 e^{x^2}\,dx$, with $f(x) = e^{x^2}$ convex on $[0,1]$, satisfies $e^{1/4} \le \int_0^1 e^{x^2}\,dx \le (1+e)/2$, i.e., approximately $1.28 \le I \le 1.86$. (Actual: ≈ 1.46.)

Jensen's inequality (integral form). The continuous analogue of the discrete Jensen: for φ convex and a non-negative weight $w \ge 0$ with $\int w = 1$,

$$\varphi\left(\int x\,w(x)\,dx\right) \le \int \varphi(x)w(x)\,dx.$$

Reverse for concave φ.

12 Limits under the integral sign

A subtle but important question: when does $\lim \int f_n = \int \lim f_n$? Pointwise convergence alone is *not enough* — the equality can fail badly.

Cautionary example. Let $f_n(x) = n\mathbb{1}_{(0,1/n]}(x)$ on $[0,1]$. Then $f_n \to 0$ pointwise on the entire interval, but $\int_0^1 f_n = 1$ for every n. So $\lim \int = 1 \ne 0 = \int \lim$.

12.1 Monotone Convergence Theorem (MCT)

Theorem (MCT)**.** *If $0 \le f_1 \le f_2 \le \cdots$ and $f_n \to f$ pointwise, then $\int f_n \to \int f$ (allowing $+\infty$).*

When to use. Pointwise-increasing non-negative sequences. Trigger: f_n obtained by truncating, by adding more terms, or by taking maxes of clean integrands.

12.2 Dominated Convergence Theorem (DCT)

Theorem (DCT)**.** *If $f_n \to f$ pointwise and there exists an integrable $g \ge 0$ with $|f_n(x)| \le g(x)$ for all n, x, then $\int f_n \to \int f$.*

When to use. Sequences not necessarily monotone, but uniformly bounded by a single integrable function (the *dominating function g*). Most flexible of the convergence theorems.

Standard example. $\lim_{n\to\infty}\int_0^1 \dfrac{x^n}{1+x}\,dx$. Here $f_n(x) = x^n/(1+x) \to 0$ for $x \in [0,1)$, $\to 1/2$ at $x = 1$. Since $|f_n| \le 1$ and $\int_0^1 1 = 1 < \infty$, DCT applies, giving the limit 0.

12.3 Bounded Convergence Theorem (BCT)

Theorem (BCT)**.** *If $f_n \to f$ pointwise on a bounded interval $[a,b]$ and $|f_n| \le M$ uniformly, then $\int_a^b f_n \to \int_a^b f$.*

A special case of DCT with $g \equiv M$.

12.4 Uniform convergence (a sufficient condition)

If $f_n \to f$ uniformly on $[a,b]$ and each f_n is continuous, then $\int_a^b f_n \to \int_a^b f$. Easier to verify than DCT but more restrictive.

13 Quick-reference cheat sheet

Pattern in the problem	Tool / trick to try first
$\int f(g(x))g'(x)\,dx$	**Substitution** $u = g(x)$
$\int u\,dv$ with u simpler after d/dx	**Integration by parts** (LIATE)
$\int \dfrac{P(x)}{Q(x)}\,dx$, rational	**Partial fractions**
$\sqrt{a^2 \pm x^2}$ in integrand	Trig substitution
Rational function of $\sin x, \cos x$	Weierstrass $t = \tan(x/2)$
$\int e^{ax}\cos(bx)\,dx$ or sin analogue	Replace by $\mathrm{Re}/\mathrm{Im}\, e^{(a+ib)x}$
$\int e^{\cos x}\cos(\sin x)\,dx$ on a period	Recognise as $\mathrm{Re}\, e^{e^{ix}}$, expand series
$\int_a^b f(x)\,dx$ with awkward f	Try $x \mapsto a+b-x$, add to original
$\int_{-a}^{a}$ with even/odd f	Reduce to $2\int_0^a$ or 0
Integrand has period T	Shift interval to convenient one
Sequence-limit $\dfrac{1}{n}\sum f(k/n)$	Recognise as Riemann sum, become $\int_0^1 f$
$\frac{1}{x^k}\int_0^x f$ as $x \to 0$	FTC + L'Hôpital
Integral with parameter, no closed form	Differentiate under integral, integrate back
$\int_0^1 (x^a - 1)/\ln x\,dx$ type	Differentiate in a, kills the $\ln x$
Mean / average value of f	MVT for integrals: $f(c) = \frac{1}{b-a}\int f$
Integral over unbounded interval	**Improper** – check convergence by p-test or comparison
Bound $\int fg$ from $\int f^2, \int g^2$	**Cauchy–Schwarz**
Compare $\int fg$ to $(\int f)(\int g)$	**Chebyshev**: same monotonicity $\Rightarrow \ge$
Convex/concave function: estimate $\int f$	**Hermite–Hadamard**
Inequality with conjugate exponents $1/p + 1/q = 1$	**Young**: $ab \le a^p/p + b^q/q$
$\lim_n \int f_n$, sequence increasing	**MCT**
$\lim_n \int f_n$, $|f_n| \le g$ integrable	**DCT**
$\lim_n \int f_n$ on bounded interval, $|f_n| \le M$	**BCT**
$\lim_n \int f_n$, $f_n \to f$ uniformly	Direct interchange
Differentiate $F(t) = \int f(x,t)\,dx$ in t	**Leibniz rule** (parametric)

Three universal moves: (1) When you see $\dfrac{1}{n}\sum f(k/n)$, immediately think *Riemann sum* — convert to $\int_0^1 f$. (2) For integrals with a parameter that resist direct evaluation, introduce or exploit the parameter and *differentiate under the integral.* (3) When trig and exponentials mix awkwardly, replace trig by Euler's formula and take real or imaginary parts at the end.